

Anish Kumar M

Agentic AI Engineer | Generative AI Developer | RAG Systems | LLM Applications

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PROFESSIONAL SUMMARY

Results-driven Agentic AI Engineer with 2+ years of experience designing, developing, and deploying intelligent enterprise AI applications using Generative AI, RAG, LLMs, LangChain, LangGraph, MCP, Python, FastAPI, and full-stack technologies. Experienced in building LLM-powered applications, Retrieval-Augmented Generation (RAG) pipelines, agentic workflows, AI tool integrations, vector search solutions, enterprise API integrations, and intelligent workflow automation. Strong hands-on expertise in prompt engineering, LLM integration, agent state and context management, Model Context Protocol (MCP), Human-in-the-Loop (HITL), structured outputs, tool calling, AI evaluation, and production deployment. Experienced in progressing from GenAI and RAG implementations to Full-Stack Agentic AI applications using LangGraph and MCP for standardized enterprise tool integration, business-process automation, marketing analytics, and operational productivity.

TECHNICAL SKILLS

Generative AI & LLM: Generative AI, Large Language Models, LLM Applications, OpenAI GPT-4o, Claude 3.5 Sonnet, Gemini, Llama 3.1, Mistral, Ollama

AI Frameworks: LangChain, LangGraph

Model Context Protocol (MCP): MCP, MCP Servers, MCP Clients, MCP Tools, MCP Resources, Tool Discovery, Tool Invocation, Context Management

Programming Languages: Python, SQL, JavaScript, TypeScript, Bash

Python Libraries: FastAPI, Pydantic, Requests, HTTPX, Pandas, NumPy, Sentence Transformers, Transformers, PyTest

RAG & Vector Databases: RAG, ChromaDB, FAISS, Pinecone, Embeddings, Semantic Search, Hybrid Search, Semantic Chunking, Metadata Filtering

Agent Architecture: LangGraph, ReAct, Sequential Workflows, Planner-Executor, Stateful Workflows, Supervisor-Worker, Human-in-the-Loop

Prompt Engineering: Prompt Templates, Few-shot Prompting, Chain-of-Thought, Prompt Chaining, Context Engineering, Structured Outputs, JSON Mode, Tool Calling Prompts

Backend & APIs: FastAPI, REST APIs, Async Python, Webhooks, Microservices, API Integration, Postman

Workflow Automation: n8n, Power Automate, Business Workflow Automation, Batch Processing, Event-Driven Workflows

Evaluation & Testing: RAGAS, Prompt Evaluation, PyTest, Response Validation, Retrieval Evaluation, Latency Testing, Accuracy Analysis

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL

DevOps & Tools: Docker, Git, GitHub Actions, Linux, VS Code, Jupyter Notebook, CI/CD

PROFESSIONAL EXPERIENCE

Agentic AI Engineer

June 2024 – Present

HCL Tech | Delhi, India

Generative AI & RAG Development

Developed Generative AI applications for digital marketing and MarTech use cases using Python, LangChain, OpenAI GPT-4o, RAG, ChromaDB, and FAISS. Built LLM-powered solutions for campaign planning, SEO content generation, customer communication, marketing knowledge retrieval, campaign analysis, and business reporting.

LLM Integration & Prompt Engineering

Integrated LLM platforms including OpenAI, Claude, Gemini, Llama, and Ollama for enterprise AI workflows. Designed reusable prompt templates, few-shot prompts, structured outputs, JSON responses, prompt chaining, and context engineering techniques to improve content quality, response consistency, and downstream processing.

RAG & Knowledge Engineering

Designed document ingestion and RAG pipelines for marketing playbooks, SEO guidelines, campaign documentation, customer personas, product information, and KPI reports. Implemented semantic chunking, embedding generation, metadata filtering, vector indexing, and semantic retrieval using ChromaDB and FAISS.

AI Tool & Python Development

Developed reusable Python tools and FastAPI services to connect AI applications with CRM systems, marketing platforms, analytics APIs, databases, and internal enterprise services. Implemented API validation, asynchronous processing, exception handling, authentication, and structured responses.

Agentic AI & LangGraph

Developed recent Agentic AI workflows using LangGraph for stateful, multi-step marketing processes. Implemented workflow state management, conditional routing, tool execution, task decomposition, and controlled AI decision flows for campaign analytics and optimization.

MCP & Enterprise Integration

Implemented Model Context Protocol (MCP) for standardized integration between Agentic AI workflows and external business tools. Developed MCP servers and MCP tools for approved CRM, analytics, reporting, and

marketing operations and integrated MCP clients with LangGraph workflows for controlled tool discovery and invocation.

Full-Stack AI Development

Developed AI-powered web interfaces using React.js, Next.js, TypeScript, and JavaScript and integrated them with FastAPI-based AI backends. Built dashboards for campaign insights, AI recommendations, workflow execution, customer analytics, and tool results.

Memory, HITL & Governance

Implemented conversation context and workflow state management for multi-turn AI interactions. Developed Human-in-the-Loop approval workflows for campaign actions, content validation, and selected business operations. Added response validation and controlled tool execution to improve AI reliability.

Evaluation, Deployment & DevOps

Developed evaluation utilities using RAGAS and PyTest to measure retrieval quality, response relevance, task completion, and latency. Containerized AI applications using Docker, managed source code with Git, and supported CI/CD deployments through GitHub Actions.

FLAGSHIP PROJECT 1: AI MARKETING AUTOMATION & GENERATIVE AI PLATFORM

Domain: Digital Marketing | **Company:** HCL Tech | **Duration:** June 2024 – June 2025

Role: Generative AI Engineer

Business Challenge

Marketing teams manually planned campaigns, generated SEO content, analyzed customer information, qualified leads, and prepared marketing reports across multiple systems. The distributed nature of marketing knowledge and repetitive manual processes increased campaign execution time and delayed decision-making.

Architecture & Technical Implementation

Developed a Generative AI-powered marketing automation platform using Python, LangChain, OpenAI GPT-4o, RAG, ChromaDB, FAISS, and FastAPI. Used LangChain as the primary GenAI framework for LLM integration, prompt management, document processing, retrieval orchestration, and RAG implementation. Developed document ingestion pipelines for marketing playbooks, SEO guidelines, campaign documentation, customer personas, and product catalogs. Implemented semantic chunking, embedding generation, metadata filtering, vector indexing, and semantic retrieval using ChromaDB and FAISS. Developed FastAPI services to integrate AI capabilities with CRM platforms, analytics systems, marketing APIs, and reporting applications.

Tech Stack: Python, LangChain, OpenAI GPT-4o, RAG, ChromaDB, FAISS, Sentence Transformers, FastAPI, PostgreSQL, Pandas, REST APIs, Docker, Git, n8n

Key Responsibilities

- Developed Generative AI applications for marketing automation.
- Built RAG pipelines for SEO and campaign knowledge retrieval.
- Implemented document ingestion and semantic chunking workflows.
- Developed embeddings and vector-search capabilities using ChromaDB and FAISS.
- Designed prompts for campaign content and marketing recommendations.
- Developed FastAPI services for enterprise AI integrations.
- Integrated CRM, analytics, and marketing APIs.
- Implemented structured JSON outputs for downstream processing.
- Developed evaluation scripts for AI-generated responses.
- Containerized AI services using Docker.

Achievements

- Reduced campaign planning effort by approximately 70%.
- Automated approximately 80% of repetitive marketing activities.
- Improved marketing knowledge retrieval relevance by approximately 35%.
- Increased campaign execution productivity by approximately 40%.
- Reduced manual reporting effort by approximately 65%.

FLAGSHIP PROJECT 2: FULL-STACK MCP MARKETING ANALYTICS AGENT

Domain: Marketing Technology (MarTech) | **Company:** HCL Tech | **Duration:** July 2025 – Present | **Role:** Agentic AI Engineer

Business Challenge

Marketing teams needed to consolidate campaign data, customer engagement metrics, website analytics, advertising performance, and business reports from multiple platforms. Existing workflows required manual data collection and separate integrations, delaying campaign optimization and operational decisions.

Architecture & Technical Implementation

Developed a Full-Stack Agentic AI Marketing Analytics platform using Python, LangGraph, MCP, FastAPI, React.js, Next.js, TypeScript, PostgreSQL, and OpenAI/Claude models. Used LangGraph as the primary Agentic AI orchestration framework to manage workflow state, conditional routing, multi-step execution, and agent decision flows. Implemented MCP as the standardized integration layer between the Agentic AI application and external marketing tools. Developed MCP servers exposing approved CRM, analytics, reporting, and campaign-management capabilities as reusable MCP tools. Implemented MCP clients to discover and invoke tools from LangGraph workflows using structured inputs and outputs. Developed FastAPI backend

services and React.js/Next.js interfaces for AI conversations, campaign dashboards, marketing insights, workflow monitoring, recommendations, and tool execution results.

Tech Stack: Python, LangGraph, MCP, MCP Servers, MCP Clients, MCP Tools, FastAPI, React.js, Next.js, TypeScript, OpenAI GPT-4o, Claude 3.5 Sonnet, PostgreSQL, REST APIs, Docker, GitHub Actions

Key Responsibilities

- Developed stateful Agentic AI workflows using LangGraph.
- Designed multi-step workflows for campaign analysis and optimization.
- Developed MCP servers exposing marketing capabilities as AI tools.
- Created MCP tools for CRM, analytics, reporting, and campaign operations.
- Implemented MCP clients for controlled tool discovery and invocation.
- Integrated MCP tool execution with LangGraph workflows.
- Developed FastAPI services for AI and MCP communication.
- Built React.js and Next.js interfaces for AI-powered marketing workflows.
- Developed dashboards for campaign insights and workflow status.
- Implemented structured tool inputs and outputs.
- Developed Human-in-the-Loop approval workflows.
- Implemented response validation and controlled tool execution.
- Containerized AI services using Docker.
- Supported GitHub Actions-based CI/CD deployment.

Achievements

- Reduced campaign analysis effort by approximately 65%.
- Automated approximately 75% of recurring reporting workflows.
- Reduced AI-to-tool integration effort by approximately 40%.
- Improved marketing insight generation efficiency by approximately 35%.
- Reduced manual marketing operations by approximately 60%.

EDUCATION

B.Tech. Artificial Intelligence and Data Science

Graduated: April 2026

Nehru Institute of Engineering and Technology • Coimbatore, Tamil Nadu

CGPA: 8.1 / 10.0